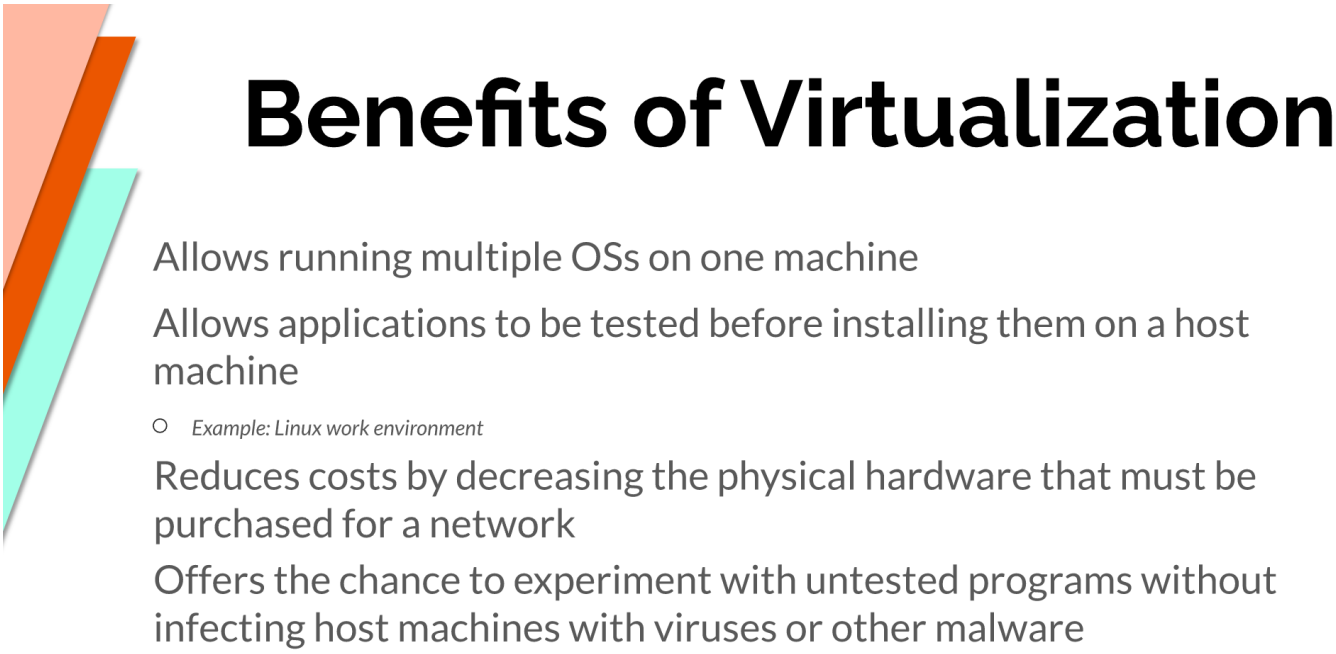
Week Report 2

Summary of Presentations

What is virtualization?

- Replication of hardware to simulate a virtual machine inside a physical machine.
- There are two general types of virtualization:
    - Server-side virtualization
        - Server-side virtualization provides a virtual desktop to each user
            - Virtual Desktop Infrastructure (VDI)
    - Client-side virtualization
        - Software installed on a computer to manage virtual machines
        - Each virtual machine has its own operating system installed
        - For client-side virtualization, the computer needs:
            - A hypervisor
                - Software that allows the management of virtual machines
            - Hardware Support
                - Capable CPU
                - Enough RAM
                - Enough storage
- Main difference between the two is where the virtualizing takes place.



Can my computer virtualize?

For a decent virtualization experience your computer should meet the following minimal specifications:

- AMD V or INTEL V compatible processor
- Dual core x64 processor with 1.3 GHz or faster
- 4GB of RAM
- Enough free hard drive space for installing guest OSs (see the minimal requirements of the desired Operating System)

Installing Ubuntu in Virtualbox

Screenshots of installing ubuntu (no more than 5)

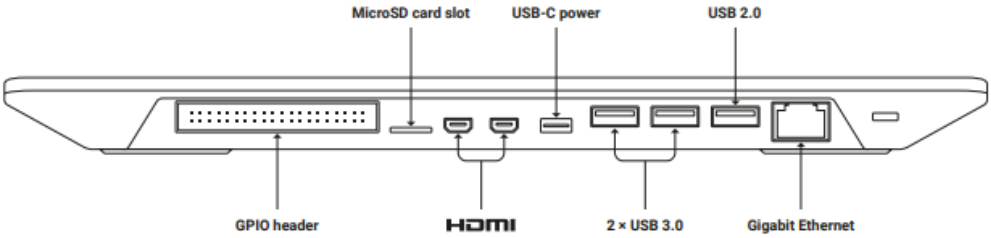
What is the Raspberry Pi?

- The Raspberry Pi is a low cost, credit-card sized computer that plugs into a computer monitor or TV, and uses a standard keyboard and mouse. It's capable of doing everything you'd expect a desktop computer to do.

Different models of the Raspberry Pi include:

- Raspberry Pi 4
- Raspberry Pi 3
- Raspberry Pi Zero W
- Raspberry Pi 3 A+
- Raspberry Pi 400

Raspberry Pi 400 Specification

Specification

Processor:	Broadcom BCM2711 quad-core Cortex-A72 (ARM v8) 64-bit SoC @ 1.8GHz
Memory:	4GB LPDDR4-3200
Connectivity:	• Dual-band (2.4GHz and 5.0GHz) IEEE 802.11b/g/n/ac wireless LAN, Bluetooth 5.0, BLE • Gigabit Ethernet • 2 × USB 3.0 and 1 × USB 2.0 ports
GPIO:	Horizontal 40-pin GPIO header
Video & sound:	2 × micro HDMI ports (supports up to 4Kp60)
Multimedia:	H.265 (4Kp60 decode); H.264 (1080p60 decode, 1080p30 encode); OpenGL ES 3.0 graphics
SD card support:	MicroSD card slot for operating system and data storage
Keyboard:	78-, 79- or 83-key compact keyboard (depending on regional variant)
Power:	5V DC via USB connector
Operating temperature:	0°C to +50°C
Dimensions:	286 mm × 122 mm × 23 mm (maximum)
Compliance:	For a full list of local and regional product approvals, please visit pip.raspberrypi.com